

|                                |                          |
|--------------------------------|--------------------------|
| Semester                       | B.E. Semester VII – CMPN |
| Subject                        | R Programming LAB        |
| Subject<br>Professor In-charge | Prof. Sanjeev Dwivedi    |
| Assisting Teachers             |                          |
| Laboratory                     | M411                     |

|                                       |                  |
|---------------------------------------|------------------|
| Student Name                          | Eshanika Amballa |
| Roll Number                           | 23102B0056       |
| Grade and Subject Teacher's Signature |                  |

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Experiment Number                                       | 06                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Experiment Title                                        | Statistical Analysis of Physical Characteristics of Palmer Penguin                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Objectives<br>(Skill Set / Knowledge Tested / Imparted) | : Descriptive Statistics, Hypothesis Testing, ANOVA & Non-Parametric Tests                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description                                             | <p>Problem Statement</p> <p>Researchers studying penguins in the Palmer Archipelago, Antarctica, collected measurements from different penguin species to understand variations in their physical characteristics.</p> <p>Using the Palmer Penguins dataset, perform a comprehensive statistical analysis in R to investigate whether physical characteristics such as body mass and flipper length differ significantly across penguin species and sex.</p> <p>The analysis should combine descriptive statistics, hypothesis</p> |

|            |                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | testing, effect-size estimation, ANOVA, post-hoc analysis, and appropriate non-parametric tests.                                                                                                                                                                                                                                                                                |
| Process    |                                                                                                                                                                                                                                                                                                                                                                                 |
| Conclusion | This analysis confirmed that penguin body mass and flipper length differ significantly across species (Gentoo being the largest) and between sexes (males heavier than females), with consistent results from t-tests, one-way and two-way ANOVA, Tukey HSD post-hoc comparisons, and Kruskal-Wallis non-parametric tests, demonstrating a robust statistical workflow using R. |

